

MINING METHODS WITH BACKFILLING IN METALLIFEROUS MINES OF INDIA

The method of extraction of ore from underground metalliferous mines is decided based on the ore body geometry, site, and geological conditions. The process involves extracting ore from large underground cavities termed as open stopes. There are two main approaches: one where the open stopes are filled with waste material (backfilling) to support the surrounding rock, and another where the rock above is allowed to collapse. In stoping with backfills, the ore body is extracted in small sections which are fully or partially filled before adjacent ore is extracted. Different open stoping methods, such as sublevel stoping, long hole stoping, and blast hole stoping, are used to extract the ore.

Presently, most of the open stoping methods are used in conjunction with backfilling (Li, 2014b). Backfilling in metalliferous mining operation is most common in steeply dipping ores bodies with restrictive dimensions and weak walls where complete extraction or selective mining is desirable. It is also applied where ground subsidence must be controlled or prevented (e.g. mining close to water logged workings, mining below townships, rivers, important surface structures etc.)

Thick and irregular ore bodies are most commonly mined in conjunction with backfill with a view to reducing losses, excessive wall closure and surface subsidence. For mining ore bodies prone to rock bursts, (like at Kolar Gold field) back-filling has been successfully applied as the filled rock mass minimizes wall closure and also absorbs the lateral stresses, and consequently backfilling helps reduce the concentration of stresses on the pillars close to the working stopes. The fill acts as a working platform in overhead cut and fill mining. The incidence of accidents due to roof falls and strata movements and rock bursts are invariably less in mines practicing backfilling.

Mining methods with backfilling in underground metalliferous mines have been developed by applying either uncemented or cemented backfills. Where ores of very high values are mined, the biggest advantages of mining with backfills are high ore recovery (90 to 100% even) and low waste dilution (less than 10%). The economic effects of these two factors can be great. Not only the life of the mine is enhanced, but the beneficiation cost gets substantially reduced due to these reasons. It is, therefore, extremely important from economic point of view that under favourable geological conditions, where open stoping could as well be adopted, backfilling

must be decided on the cost-benefit analysis. Decisive factors in choosing the right filling method are the quality and cost of filling materials.

Mining with backfills was introduced in Indian metalliferous mines for the first time in the year 1953 at Kolar Gold Field. Since then, several mines have been using any one or more of the following mining methods applying either uncemented or cemented backfills (Shetigar. 1983):

- i) Breast stoping with hydraulic filling for ore bodies less than 1.5 m in width;
- ii) Room & Pillar stoping with backfilling for ore bodies less than 6.0 m width and not more than 80 m along strike;
- iii) Cut and fill stoping for ore bodies exceeding 4.0 m in width and a minimum strike length of 80 m;
- iv) Post-pillar stoping for ore bodies exceeding 6.0 m in width and a minimum strike length of 80 m.
- v) Sub-level stoping with delayed mass filling
- vi) Long hole stoping with backfilling

A few underground metalliferous mining methods with backfilling are briefly explained hereafter.

8.2.1 Breast Stopping with Backfilling

In the breast stoping method, a raise is initially put up along the hanging wall of the ore body dipping at about 30° between two successive levels with a vertical interval of 37.5m. Holes of lengths 2.0 m and 32 mm dia. are drilled on each face at an angle of 45 -50° and about 60-80 numbers of drill holes are blasted electrically at a time. Both faces of the raise are advanced in the strike direction leaving crown and sill pillars systematically and also chute pillars for protection of levels. Random stope pillars are left in geologically disturbed areas. The excavated area is supported by 200 mm dia. timbers erected at intervals of 1.5 m along the dip and 2.4 m along the strike (Fig.8.8). In addition, 60 cm chock-mats are erected at 4.5 m intervals as an additional support to the hanging wall. An electrically operated 22.5 kw double drum slusher installed on the lower horizon, brings down the broken ore into the timber chute, from where it is filled in 1.4 tonne side-tipping mine-cars. After completion of stoping in a block,

strong barricades are fitted in the chute raises and backfilling with deslimed mill tailings is done.

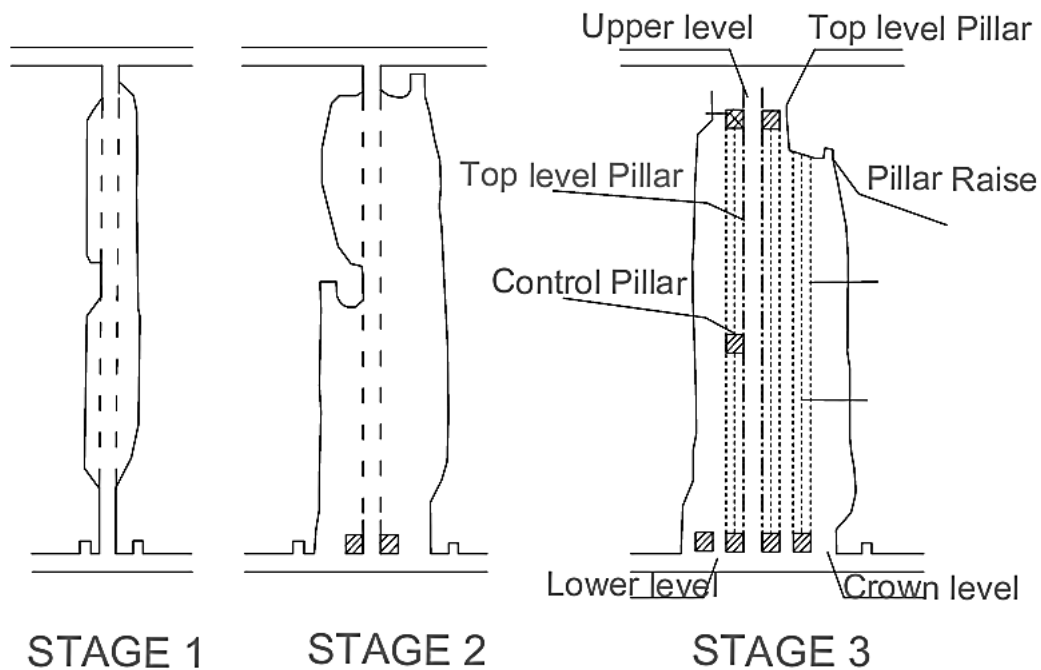


Fig. 8.8. Line diagram showing breast stoping (after Shetigar, 1983)

8.2.2 Room and Pillar Stopping with Backfilling

In Room and Pillar Stopping (Fig.8.9) a raise is initially mined along the hanging wall contact of the ore body from lower horizon and both the faces of the raise are advanced to a span of 15 m leaving 4.0 m wide rib pillars between the adjacent rooms. Back is supported by systematic tor-steel bolts 1.5 m long and 20 mm in dia at a spacing of 1.2 m x 1.2 m. After completing the hanging wall cut for a height of 2.0 m, the ore left in the stope is stripped out by drilling 2.0-2.4m long holes. In case of wider ore body, floor stripping and scrapping are repeated. Systematic chute and level pillars are left to protect the levels. After stoping operation is over strong barricades are erected at the bottom and backfilling of the room is done. Ore production from a Room & Pillar stope normally varies from 1,000-2,200 tonnes/month and maximum may reach 3,000 tonnes/month.

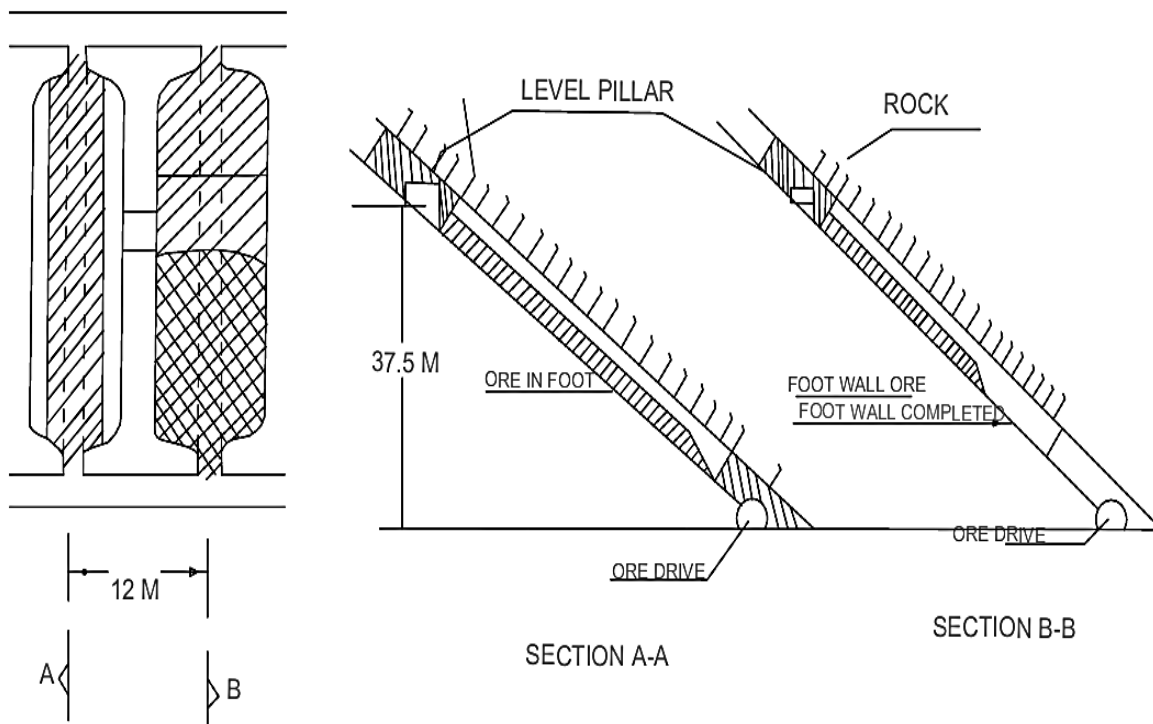


Fig. 8.9. Layout of room and pillar stope (after Shetigar, 1983)

8.2.3 Cut-and-Fill Stopping

Cut and fill stopping is generally adopted in veins and inclined tabular ore bodies, where the host rock mass is of insufficient strength to sustain the significant unsupported spans. Brady and Brown (1993) suggested that this method is suitable for ore bodies with dips in the range of 35-90°, while the stope spans may range from 4.0-40 m, though 10-12 m usually the upper limit. In cut and fill stopping, mining advances in the bottom-up sequence, whereby an initial ore drive is developed at the bottom of a stope block. Backfill is then placed on the floor of the drive to act as a platform for developing another ore drive, as illustrated in Fig. 8.10. In contrast, the top-down mining sequence requires horizontal crown pillars to act as a barrier against the inrush of uncemented backfill from the overlying mined-out stopes. They also act as regional pillars for the long-term stability of the mine structures.

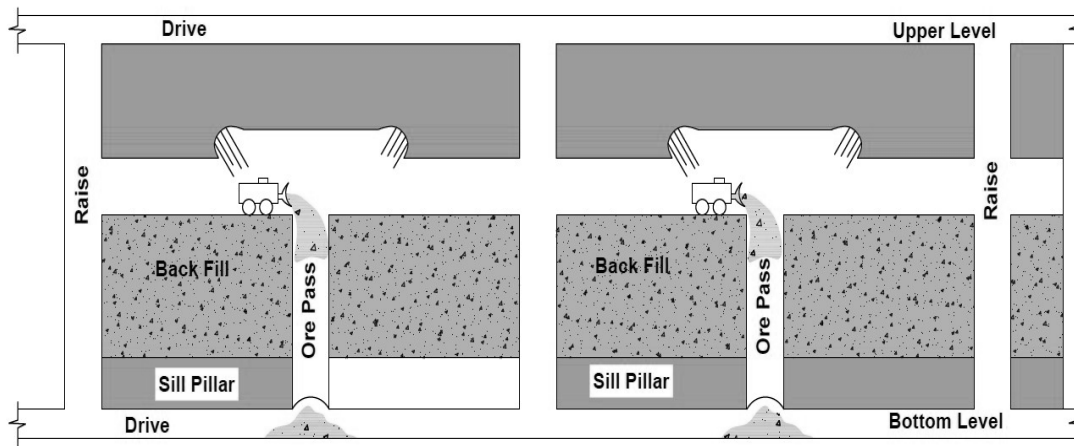


Fig. 8.10. A typical cut and fill stoping (Overhand)

In this method of stoping, raises are driven between two levels. Sufficient sill pillar is left above lower level to support lower level. A sill drive is opened in the region above sill pillar, to open the stope, heightening of sill drive is further done as a part of extraction of material up to designed height. Excavated part is now backfilled with either cemented or uncemented backfill depending on site condition to work as platform for further excavation. This cut and fill operation propagates towards upper level. Crown pillar is left below upper level to support it. Ore passes are made which may be equipped with chute at the bottom to transfer material at loading level by gravity. Design of sill pillar and crown pillar depends on rock mass strength parameters of rock and accordingly stress analysis due to excavation. Raises serve the purpose for ventilation and sometimes it also may be equipped to serve as manways. Generally, a stope is worked in two panels so that a loader can be deployed in both the panels alternately. An LHD is deployed to load and haul the broken ore to the ore passes. As preparation for backfilling is made in one panel, the ore is extracted from the other panel adhering to a rigid cycle of operation. A production from horizontal cut and fill stope in a mine may vary from 2,500-3,500 tonnes/month. Diagrammatic layout of Cut and Fill stoping method is shown in Fig 8.11.

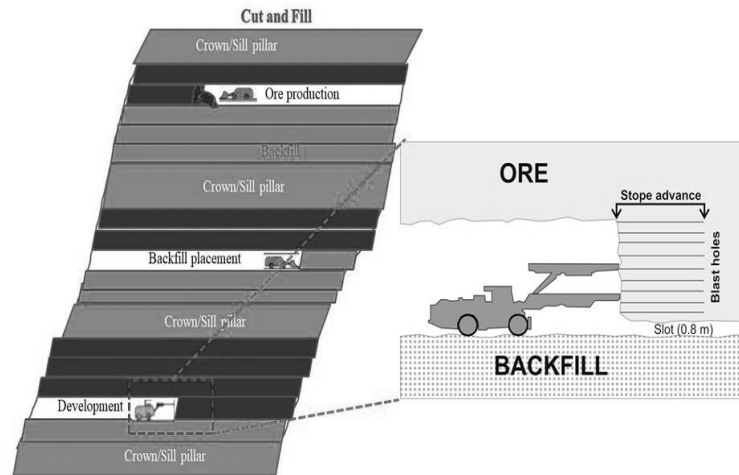


Fig. 8.11. Schematic of a typical cut and fill mining method (modified after Abzalov, 2016).

Horizontal cut-and-fill method with a little variation in the drainage system as practiced at Jaduguda Uranium mines, is illustrated in Fig. 8.12. Mill tailings of 50-60% consistency is pumped through 75mm dia pipes into the stopes (Dhar et al, 1983).

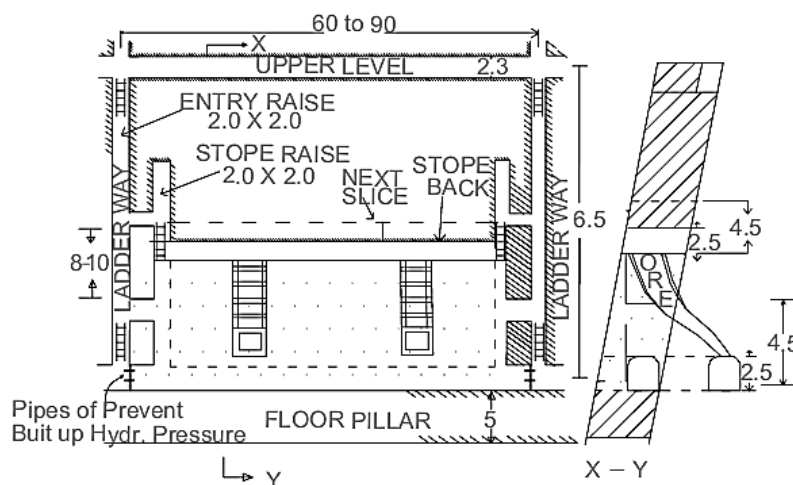


Fig. 8.12. Layout of cut-and-fill practice at Jaduguda Uranium mine (after Dhar et. al. 1983)

8.2.4 Post Pillar Stopping with Backfilling

Post pillars stopping with backfilling method was introduced at Mosabani Group of Copper mines in 1973 as well as for extracting thick Pb-Zn ore bodies of Rajpura Dariba mines in

district Udaipur. It is similar to horizontal cut-and-fill stoping as explained above, except for the formation of 4.0 m x 4.0 m in-situ vertical pillars, called "Post-Pillars", which provided additional stability to the roof and permit wider span to be excavated. In between the post pillars a clear space of 13 m along the strike and 9.0 m across it was left at Mosaboni Group of Mines (Fig.8.13).

However, at Rajpura-Dariba mine, where a 24m wide ore body was under extraction by this method, the pillar size was 6.0 m x 6.0 m placed at intervals of 20 m center-to-center both along the strike as well as across it leaving a clear span of 14 m. Since the hanging wall and backs at both the mines were highly disturbed with a number of joints, slips and fractures, cable bolting was done by drilling 57 mm dia. and 18 m long holes at a spacing of 3.0 m x 2.4 m and cement grouting after placing 16.2 mm dia. high tensile steel cables.

Stoping operation by post pillar method, starts with the development and full exposure of the lode from hanging wall to the footwall contact. A free face is then created near the hanging wall contact so that drilling can proceed systematically towards the footwall side. A small stretch of the cut on the footwall is separately stripped at the last stage of the drilling cycle. In Mosaboni mines, normally, 2.2 m high cuts were taken by drilling 2.3 m long upper holes at 70° to the horizontal, using wagon drills and the height of the excavation is limited to a maximum of 4.8 m above the backfill. However, at Rajpura Dariba mines, the ore was extracted in slices of 3.0 m. The broken ore was loaded by LHD in both the mines (0.76-1.67 m³ capacity at Mosaboni) and it was dumped into the nearby ore passes. The conventional wooden chutes fitted in the ore passes were later replaced by manually operated steel with a view to handling larger tonnages and speeding up of the loading of mine-cars. Soon after the entire ore was cleared from the stope, the ore passes were raised using 1.5 m dia. steel rings and backfilling was completed in a panel leaving a head room of 2.4 m. While deslimed mill tailings alone were used in Mosaboni Copper Mines, a mixture of deslimed tailings and sand mixed in the ratio of 1:3 by weight was used at Rajpura Dariba mines (Thakur et. al 1992). A minimum of 2 nos. of ore passes per extraction panel were considered essential for efficient loading. On an average 3000-4500 tonnes/month of output was obtained from each of post pillar stopes in Mosaboni area whereas after introduction, sand-mill tailings mixture at RD mines, the output from the mine was increased from 1000-1200 tonne/day to nearly 1500-1800 tonne/day (Rathore 1992)

Efficient drainage arrangement of water from the poured-in-fill is an essence of success in all overhand cut-and-fill mining systems. In Mosaboni mines, the rate of percolation of water was found to be getting reduced after extraction of 3-4 cuts and backfilling and due to this the cycle of operation was getting delayed. This problem was overcome by drilling NX size boreholes through the sill pillar and by fixing old perforated pipes wrapped with wire mesh and hessian cloth and extending it before pouring fresh fills.

Similar perforated pipes were laid after every 2-3 cuts on the floor of the backfill along the strike direction and also across it (in case of wide ore bodies) to ensure faster drainage of water from all parts of the fill. At RD mines, drainage from the fill got improved after introducing tower drainage pipes, constructed by cutting the normal pipe into two halves and welding semicircular wire mesh to it and wrapping it with hessian cloth before fixing in the stopes. Addition of sand to partially deslimed tailings at RD mines further improved the rate of percolation through these tower pipes and persons could walk over the fill during backfilling.

In both the mines, dilution of the blasted ore along with backfill, was a serious problem affecting the mill operation. Although lot of care was taken during the operation of the LHD to avoid too much of digging into the backfill, the problem still remained unresolved. Decision on introduction of cemented fill layer was getting delayed due to high cost of Portland cement. Blasting over wooden planks laid over the backfill was found to be a cumbersome operation. Apart from these, low velocity of air in the post-pillar stope due to large area of exposure was another problem being faced at Mosaboni mines. In spite of the above problems, post-pillar stoping with backfilling was widely practiced in Mosaboni Group of mines for mining wide ore bodies under adverse geological conditions and along with horizontal-cut-and-fill method accounted for nearly 70% of the total output from Singhbhum Copper Belt. Compared to an O.M.S. of nearly 1.5 tonnes with breast-stoping, an O.M.S of 3.5- 5.5 tonnes was obtained in post-pillar stopes. The cost per tonne of ore produced by post-pillar stoping was nearly 50% that with breast-stoping and nearly 60% that compared to room & pillar mining (Singh, R.D and Singh, D.P.,1983). Post-pillar-mining gave a high percentage extraction of about 85% (Dhar, B.B, et al., 1983).

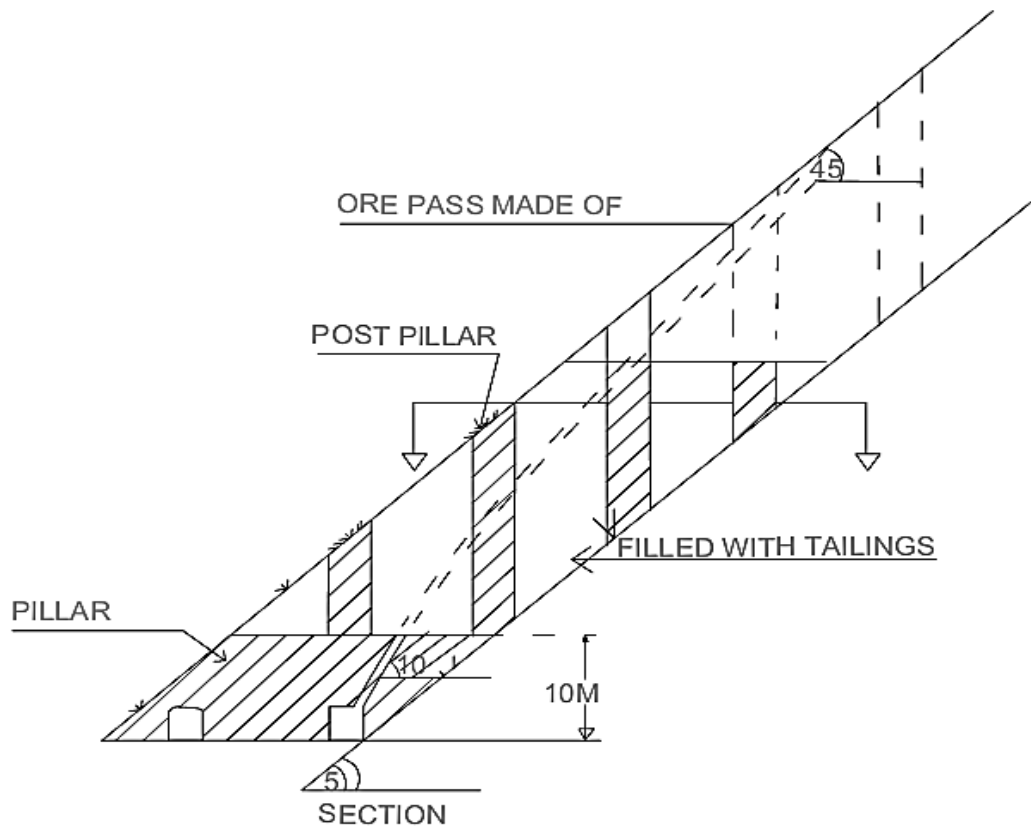
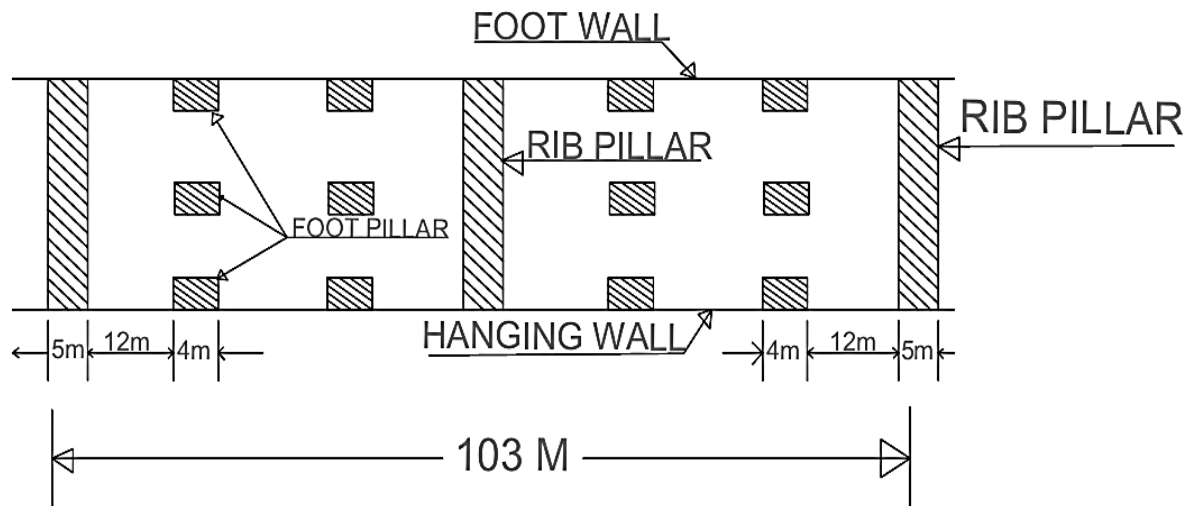


Fig 8.13: Layout of Post Pillar Stope (after Shetigar, 1983)

8.2.5 Sub-level stoping with delayed mass filling

The sublevel stoping method is typically used under geological conditions, such as steep ore bodies, stable hang wall (H/W) and foot wall (F/W), competent host rock and ore body, regular ore deposits. The open stopes, also termed as blasthole stopes, created after the extraction of

ore in sublevel stoping are usually backfilled to allow maximum recovery from the stopes (Hartman and Mutmanský, 2007). The schematic layout of sublevel stoping is illustrated in Fig. 8.14a. The ore body is divided into blocks by driving levels at different depths. The levels are connected to the surface via ramps, inclines/declines/shafts so that men, materials, and machineries can move from underground to the surface, and vice versa (Deshmukh, 2009; Villaescusa, 2014, Villaescusa et al., 2019). The sublevels are driven by creating raises in an upward direction on either side or through the middle of the ore body. In between the levels, sublevels are made depending on the production requirement and stability of the underground workings. The ore body is developed to form separate stopes, between which the ore is left intact in pillars to serve as a natural support to the H/W and roof. Horizontal slices of the ore are also left along the top of the stopes, termed as crown pillar. The blasted ore is collected at the bottom level cross cuts, termed as draw points (Sletten, 2012).

In sublevel stoping with backfilling, the extracted stope is filled with suitable backfill materials. In case of the ore bodies with considerable strike length and thickness, the ore body is divided into different stopes called primary and secondary stopes. Generally, in secondary stopes, the ore is extracted after the backfilling of the primary stopes (Fig. 8.14b). It also permits simultaneous mining in different stopes; wherein multiple stopes are mined simultaneously in both directions from the center of the ore body.

In India, sub-level stoping with delayed mass filling was introduced for the first time at Kalyadih Copper Mine, Karnataka for mining a 30 m wide chalcopyrite ore body. The hydraulic fill plant was designed by Thakur et al (1986). Fresh mill tailing was deslimed by a hydro-cyclone to contain 8-12% of -53micron fractions and was pumped to an overhead bin fitted with a central pipe and rubber plug with tangential holes. By manually raising the rubber plug and injecting water therein, the semisolid slurry from the bin flew down into a gravity mixer where additional water was added, whenever required. Arrangements had been incorporated at the ground level to mix 100 tonnes of deslimed tailings with 50 tonnes of unclassified tailings per day from the old stock to compensate for the loss during desliming. The ensued slurry was passed first through a borehole pipe and then along drives to the top level of the stope from where it was released into the stope.

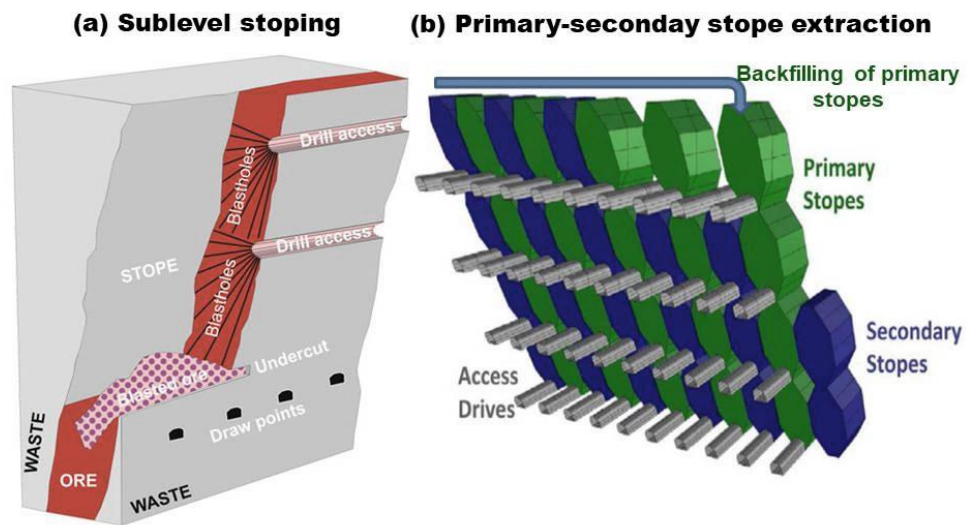
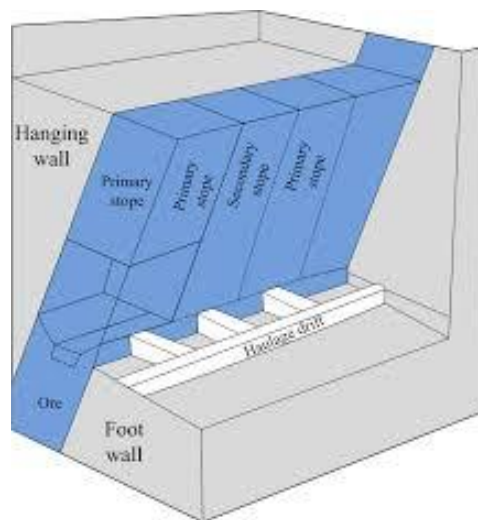
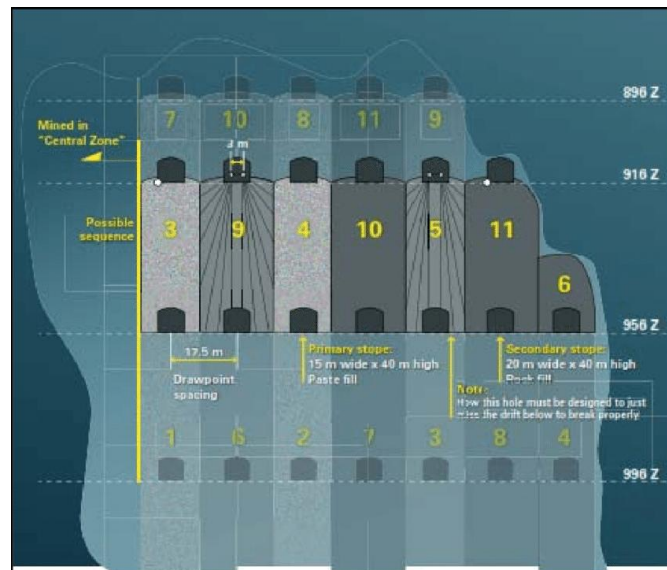


Fig. 8.14 Sublevel stoping method: (a) Schematic diagram (modified after Abzalov, 2016), and (b) Primary-secondary stoping extraction (modified after Sivakugan et al., 2015).





8.2.6 Long hole stoping with backfilling

In massive, regular, and large ore bodies, selective mining may not be appropriate from the cost perspective. Large size underground mining machineries and bulk mining method like long hole stoping may be used under such circumstances. Hence, a large volume of ore can be extracted with high productivity (Abzalov, 2016). Long hole stoping, a variant of sublevel open stoping, is a highly productive and low-cost metalliferous mining method in which large diameter (140 to 165 mm) blast holes are used. The depth of the blast holes may reach up to 100 m. In this method, the miners create a vertical slot at one end of the stope and then work in the levels to drill a radial pattern of drill holes (Gonen and Kose, 2011). After charging a set of these blast holes with explosives, the blocks of ore body are blasted. After extraction of the stopes, the primary voids are backfilled with cemented backfill, and the secondary voids are backfilled with waste rock (uncemented), as shown in Fig. 8.15.

There are two main variants of the long hole stoping method: Transverse and Longitudinal stoping. If the ore body is mined by retreating from the hang wall (H/W) to the foot wall (F/W) for thick ore bodies, it is referred to as transverse stoping. Here, the long axis of the stopes and drifts are perpendicular to the strike of the ore body. In contrast, extraction from stopes along the strike of the ore body is referred to as longitudinal stoping. This approach is used if the thickness of the ore body is lesser and is not suitable for transverse stoping.

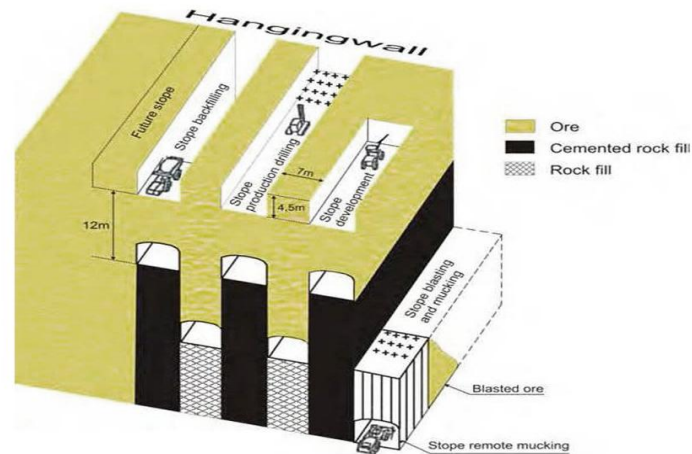


Fig. 8.15 Long hole stoping with backfilling (Gonen and Kose, 2011).

Underhand mining with paste fill

Overhand (bottom-up) mining is a traditional mining method used for extracting ore in orebodies dipping at 50° or greater. In this method, miners work on previously backfilled stopes and extract ore in an upward direction in the form of benches, with the broken ore falling to a lower level for removal. This method performs well in stable ground conditions with competent wall rocks and where the roof is unlikely to collapse. However, it requires initial development to access deeper levels, which can be time-consuming, capital-intensive, and may delay the start of ore extraction.

Advantages

- Suitable for:
 - Stable ground conditions
 - Strong wall rocks
- Simple and widely used method

Limitations

- Requires extensive initial development
- Time-consuming and capital-intensive
- Delays start of production
- Requires crown and sill pillars

Issues with Overhand Mining

- Pillars contain valuable ore: loss of resources
- Difficult to extract pillars at greater depths
- High stresses cause:
 - Ground failure
 - Unsafe conditions
- Poor geology (faults, shear zones) worsens stability

In conventional overhand mining, crown and sill pillars are left to separate and support upper and lower levels of the mine. These pillars often contain valuable high-grade ore, which may be difficult to recover and can be permanently lost unless extracted. As mining depth increases, extraction of these pillars becomes more challenging due to higher stresses. Poor ground conditions further aggravate the situation, leading to ground failures and unsafe mining conditions. One effective solution is to adopt an underhand mining method using cemented paste fill (CPF) to stabilize the overlying voids.

The underhand mining method is used for safe ore extraction in conditions characterized by complex geology and high stress. In this method, mining progresses in a top-down sequence, eliminating the need for crown and sill pillars, as extracted stopes are backfilled with CPF. For successful implementation, CPF must achieve sufficient strength to allow safe development and prevent failure of the fill roof when exposed during the extraction of underlying stopes.

Key Features

- Mining progresses top-down
- No need for crown/sill pillars
- Stopes are backfilled with CPF

Advantages

- Improved safety
- Better ore recovery
- Suitable for high-stress environments

Factors Affecting Stability

- Cement content
- Fill height
- Geometry of stope
- Curing time

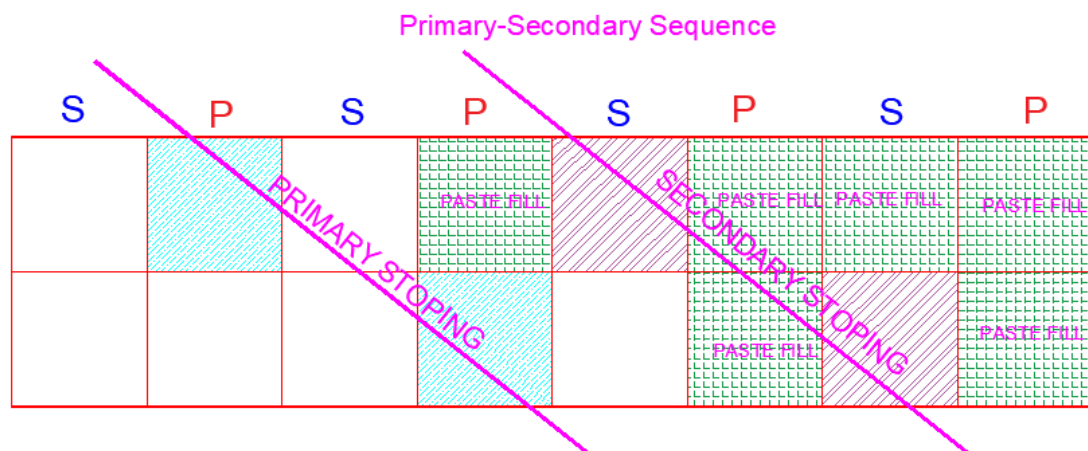
Instrumentation

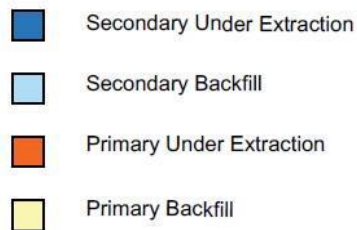
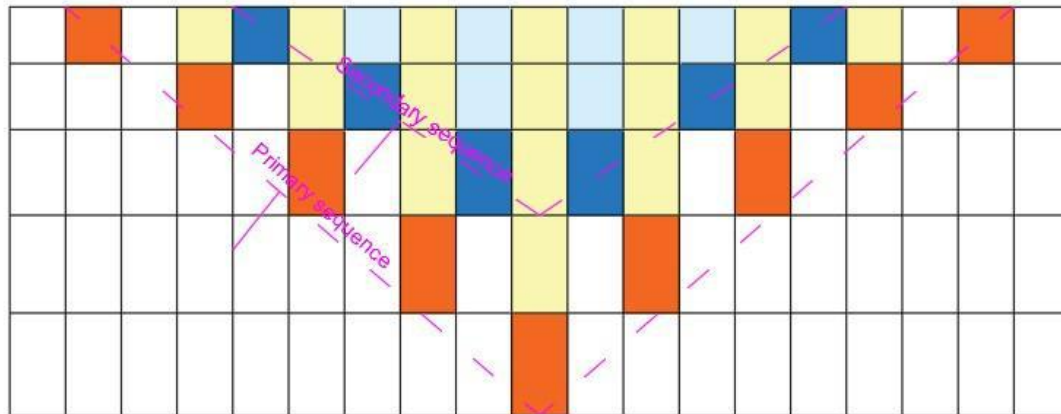
- Tell-tales : measure displacement
- MPBX : monitor deformation
- CMS: analyze stope geometry

The underhand mining method was first implemented in the Star and Lucky Friday mines through collaboration between the US Bureau of Mines, NIOSH, and Hecla Mining Company. This approach was developed to improve safety in rockburst-prone conditions. The stability of drives constructed within backfilled stopes is crucial, as they must remain stable until the excavation of underlying stopes is completed

Primary-secondary Sequence of stoping

In primary-secondary sequence, a double diagonal line of extraction is maintained, with the primary stoping at a lead of 2 stopes with respect to the secondary stoping in the same sub-level. The bottom level will have the similar primary-secondary sequence with a lag of 1 or 2 stopes with respect to the top level as shown. Thus, a double diagonal line of extraction is maintained

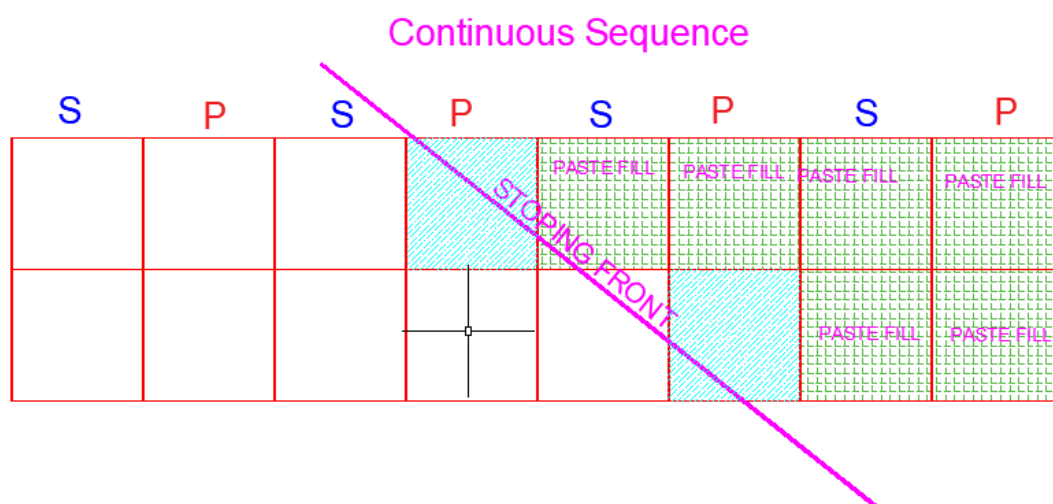




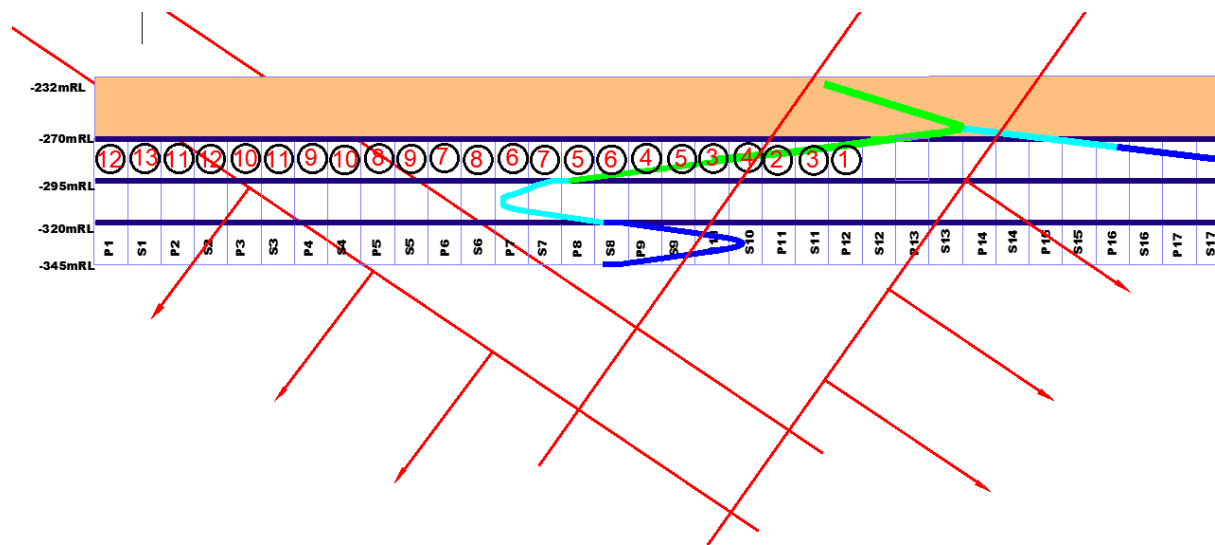
Underhand sequence of mining with paste filling required re-development of roadway in paste backfill.

Continuous sequence of stoping

In continuous sequence, the stopes are extracted continuously, by keeping the bottom level at a lag of 2 or 3 stopes with respect to the top level



Proposed Mining sequence to reduce the stress damage in FWD and enable crown pillar recovery.



Example:

Geology of the Mine Site

The orebody is situated in the central region of the eastern limb of a prominent synformal fold. The dominant rock formations in this area comprise interbedded mica schist, chert, and quartzite, which together form a ridge trending in the NNE–SSW direction.

The mineralization of lead, zinc, and silver is primarily associated with calc-silicate bearing dolomite and graphite mica schist. Additionally, graphite mica schist and calcareous quartz-biotite schist are exposed on the eastern side of the ridge.

The region exhibits significant structural complexity due to the presence of major shear zones, extensive regional faults, and other geological discontinuities, as depicted in Fig. 1. The orebody generally has a thickness of around 30 m and follows a strike direction between N10°E and N15°E.

At shallower levels, the orebody dips westward at angles ranging from 45° to 60°. However, at depths beyond 400–500 m, structural folding alters the dip direction to the east, with steeper angles between 70° and 80°.

This geological complexity necessitates the adoption of underhand mining techniques for safe and efficient extraction. The quality of the rock mass is indicated by Rock Quality Designation

(RQD) values ranging from 60 to 99 and Rock Mass Rating (RMR) values between 43 and 67 for both ore and surrounding waste rocks.

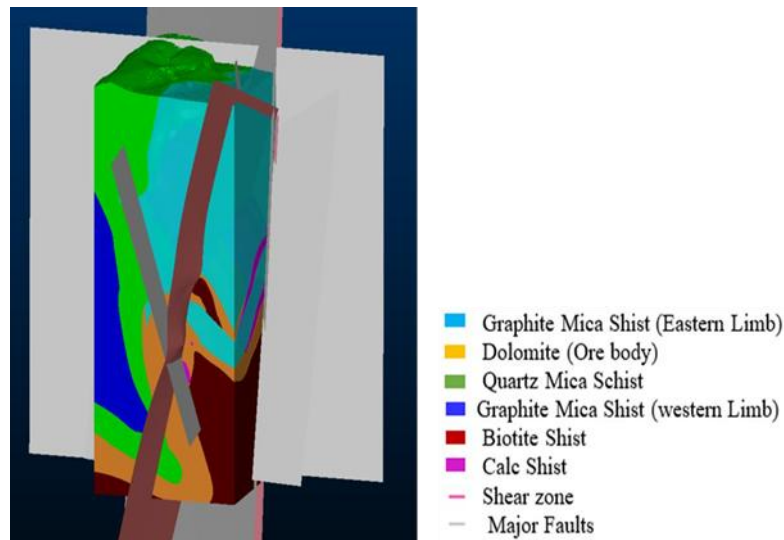


Fig. 1 3D Representation of lithologic structural model

Method

At this mine, underhand mining is implemented at depths exceeding 800 m for ore extraction. The operational sequence is illustrated in Fig. 2a–2d. Once a stope is excavated, it is backfilled using Cemented Paste Fill (CPF) in two distinct stages.

In the initial stage, referred to as plug filling, the CPF mixture consists of approximately 10% cement and 90% mill tailings, with a water content of about 22%. This material is placed up to a height of 9 m from the extraction level. To contain the fill material and maintain structural integrity, the extraction crosscuts are sealed with robust shotcrete barricades.

After sufficient curing of the plug fill, the cement proportion is reduced to about 4%, and the remaining volume of the stope is filled. This second phase is known as bulk filling, which continues up to the drill level.

Following a curing period of roughly 28 days, a drill drive is developed through the plug fill using drilling and blasting techniques, typically employing a burn-cut pattern (Fig. 2a). The drive has dimensions of 4.8 m in width and 4.8 m in height. In this configuration, the floor consists of competent ore, whereas the roof and sidewalls are formed by the cemented paste fill.

The developed drill level facilitates production drilling and charging operations within the stope (Fig. 2b). The blasted ore is subsequently removed from the extraction level (Fig. 2c). Once extraction is complete, the stope is refilled (Fig. 2d), and the cycle is repeated for the next lower stope.

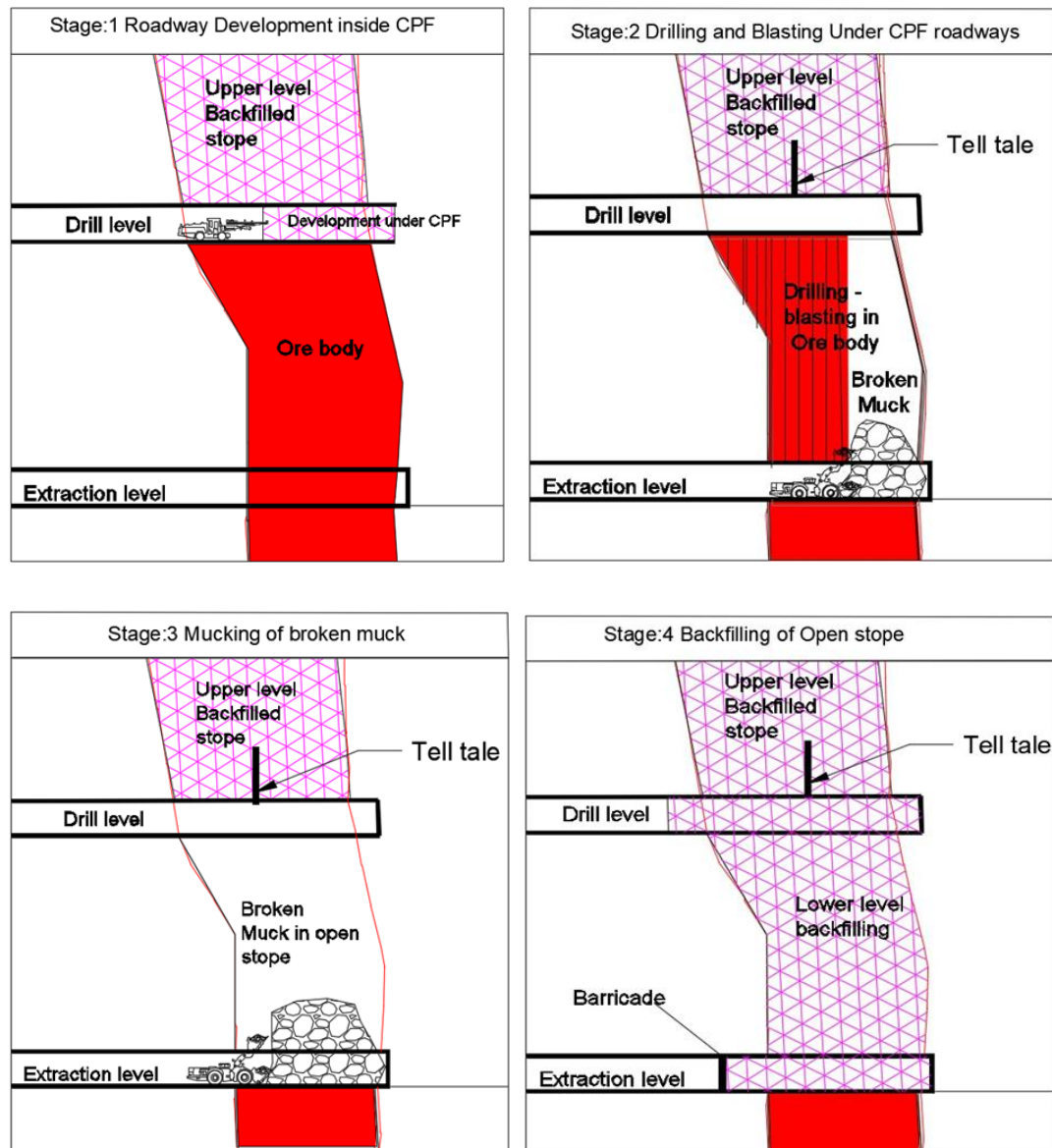


Fig. 2 Sequence of underhand mining method using paste back fill

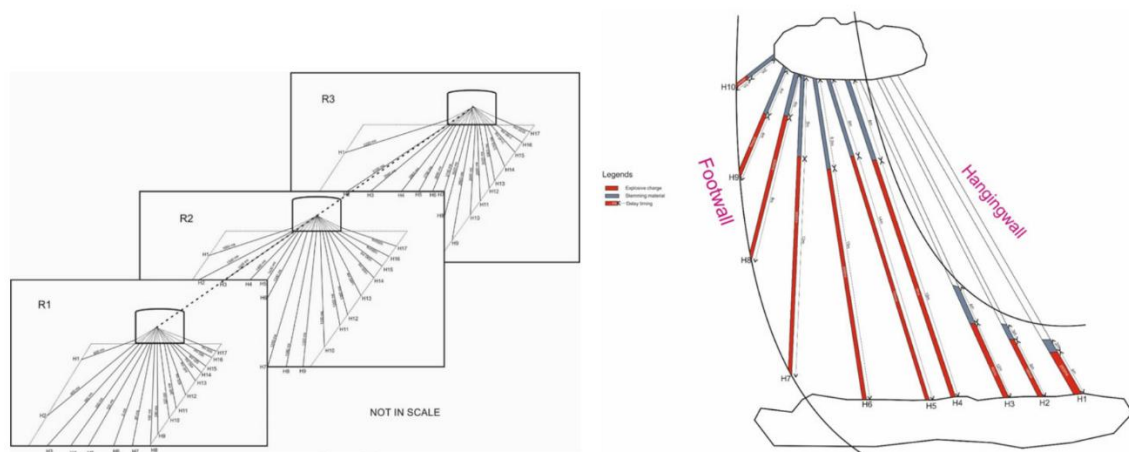
Ground Support System

After the excavation of the drive within the CPF, appropriate ground support measures are implemented to ensure stability and safety. Initially, a 50 mm thick layer of shotcrete is applied to the surface. This is followed by the installation of wire mesh, which is anchored using

friction bolts measuring 2.4 m in length and 48 mm in diameter. These bolts are installed at a uniform spacing of $1 \text{ m} \times 1 \text{ m}$ in both longitudinal and circumferential directions.

A second layer of 50 mm thick shotcrete is then applied to complete the support system. This combined arrangement of shotcrete, wire mesh, and friction bolts effectively stabilizes the CPF drive and provides safe working conditions for personnel .

Production drill and blast



Multi ring blasting

Ring blasting

- Larger diameter drilling and blasting
- DTH
- Hole depth: Depends on stope height
- Drilling in advance of blasting face.
- Explosives
 - Cartridge emulsion/Slurry
 - Site mixed emulsion (SME)/site mixed slurry (SMS)
 - Charging down hole or up hole or combination of both
 - Compressed ANFO used for up hole charging, compression provides adhesiveness to explosive to remain within blast hole.
 - High viscous SME/SMS for up hole charging
 - Ease in pouring downhole
 - Ease in pumping up hole.
- Ring blasting pattern

